FOR COMPLETE BEGINNERS

100 lines or 3?

Complete Modern Java, episode 12: loops, in Java and TypeScript

> A loop repeats work with three parts: start, keep going while, move on.

A countdown by hand is four lines; from a hundred, a hundred and one.

A loop repeats the work. Forget the move-on part, and it never ends.

In this episode

- while loops
- The infinite loop and Ctrl+C
- for loops and i++
- 0 to < 3
- Running totals with +=

Coming next

- Methods (episode 13)
- Arrays and lists (later)

At the time of recording: Java 25, Node 24 LTS, IntelliJ IDEA 2026.1.3.

01 What each loop printed

while (count > 0), count from 3	`3 2 1 Go!`
while with no update	`never finishes (Ctrl+C)`
for (int i = 1; i <= 3; i++)	`Item 1 Item 2 Item 3`
for (int i = 0; i < 3; i++)	`0 1 2`
three loaves, totalCents += 150	`450`
TypeScript: the same two loops	`3 2 1 Go! Item 1 Item 2 Item 3`

Every row was run, not predicted.

02 while and for

The check runs before every repetition. False at the start? The body never runs.

for puts start, check and update on one line. i++ adds one after each repetition.

Countdown.java and Repeat.java

```
int count = 3;
while (count > 0) {
    IO.println(count);
    count = count - 1;
}
for (int i = 1; i <= 3; i++) {
    IO.println("Item " + i);
}
```

> Printed 3 2 1 Go!, then Item 1, Item 2, Item 3.
>
> Run on a Mac, 2026-10-24: Java 25.0.4.1, Node 24.22.1, TypeScript 7.0.2.

03 A running total

Start at 0 and add each time round. += adds to the variable.

Loaves.java

```
int totalCents = 0;
int breadPriceCents = 150;
for (int i = 1; i <= 3; i++) {
    totalCents += breadPriceCents;
}
IO.println(totalCents);   // 450
```

> Printed 450.
>
> Run on a Mac, 2026-10-24: Java 25.0.4.1, Node 24.22.1, TypeScript 7.0.2.

Before you run a loop

1. Where does it start?
2. When does it keep going?
3. What moves it towards the end?
4. Stuck? Ctrl+C in the terminal.
5. Challenge: predict for (int i = 2; i <= 6; i += 2).

Everything else

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- **The code and the challenge files: complete-modern-java**
  https://github.com/code-with-sam-dev/complete-modern-java

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